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f26.207.quiz05::30

In all quizzes and exams, show your work and write with correct notation.

(10pts) 1. Given $f(x) = x^3$, find an equation to the tangent line of $y = f(x)$ at $x = 2$

$f'(x) = 3x^2$

$f'(2) = 3(2)^2 = 12$

$f(2) = 2^3 = 8$

$y - 8 = 12(x - 2)$

+0 optional

(10pts) 2. Given $f(x) = x^6 \cdot \dfrac{8^x + 1}{x^2 + x}$ find $f'(x)$

Parentheses needed due to Order of Operations

parens parens parens

$$f'(x) = x^6 \cdot \frac{(x^2+x)(8^x \ln 8 + 0) - (8^x+1)(2x+1)}{(x^2+x)^2} + \frac{8^x+1}{x^2+x} \cdot 6x^5$$

This can be directly differentiated by Quotient Rule:

$$\left(\frac{1}{x^2}\right)' = \frac{(x^2)(0) - (1)(2x)}{(x^2)^2}$$

(10pts) 3. Given $f(x) = \sqrt{x} + \dfrac{1}{x^2}$ find $f'(4)$

$f(x) = x^{1/2} + x^{-2}$ ← still $f(x)$ not $f'(x)$

⚠ $f(x) \neq f'(x)$

$f'(x) = \frac{1}{2}x^{-1/2} + (-2)x^{-3}$ this can also be $-\frac{2}{x^3}$

$f'(4) = \frac{1}{2}(4)^{-1/2} + (-2)(4)^{-3} = \frac{1}{2} \cdot \frac{1}{4} - \frac{2}{64} = \frac{1}{4} - \frac{1}{32} = \frac{7}{32}$

You can leave it like this

Do not mix terms of $f(x)$ and $f'(x)$ Example: $x^{1/2} + (-2)x^{-3}$ (!)